Jeffrey Hoelzel Jr.

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AI Engineer | Full Stack & AI Systems

AI Engineer at VIAVI Solutions building production software across full-stack applications, MCP servers, developer tools, and AI agents. Experienced designing AI-enabled backend services, agentic LLM workflows, cloud-deployed applications, and data pipelines for healthcare, research, and enterprise use cases. Strong foundation in Python, TypeScript, APIs, and scalable systems.

Education

Northern Arizona University, Flagstaff, AZ **GPA: 3.86**

B.S. in Software Engineering | Minor in Mathematics Dean's List

Awards: Outstanding Senior Award | Keim Undergraduate Research Excellence Award, Honorable Mention | HURA Grant Recipient

Work Experience

AI Engineer I – R&D Rotational Program | VIAVI Solutions | Colorado Springs, CO **Jul 2026 – Present**

- Cut SQL query MCP tool memory use ~230x by removing redundant parse step, eliminating a reliable crash on large queries.
- Designed a proactive memory watchdog for sandbox code execution, cutting kills of unrelated jobs from 67% to 0%.
- Shipped a token-based hosted-link mechanism so AI-generated artifacts can be rendered safely in any chat client without new client-side support.

Software Engineer Intern | Altored Health | Flagstaff, AZ **Jan 2026 – Jun 2026**

- Architected semantic retrieval workflows using pgvector and Meilisearch to match ambiguous patient queries with structured healthcare visit targets.
- Implemented confidence-gated LLM fallback logic with strict JSON outputs, increasing correct target selection from 67% to 98%.
- Dockerized backend services using ACR and deployed the chatbot as an Azure Container App microservice to support scalable API orchestration.

Machine Learning Engineer | Pathogen and Microbiome Institute | Flagstaff, AZ **Aug 2025 – Aug 2026**

- Developed PepSeqPred, a PyTorch-based ML pipeline to predict antibody epitope locations across diverse pathogens.
- Optimized ESM-2 embedding generation pipelines using SLURM GPU parallelization, reducing HPC runtime by over 50%.
- Built logging and evaluation tools using PR AUC, recall, precision, F1, and confusion matrices to validate model performance under extreme class imbalance.
- Developed a C++ genomics pipeline with Bash shell jobs for adapter trimming, genome assembly, polishing, and circularization.

Information Technology Intern | Cavco Industries | Phoenix, AZ **Jun 2025 – Aug 2025**

- Built agentic LLM workflows to classify, route, and update ServiceDesk Plus ticket operations through automated tool calls.
- Designed a RAG pipeline, reducing response latency by 60% and consolidating 3+ API calls into 1 request.
- Developed an ML pipeline clustering 100,000+ tickets to automatically generate Confluence knowledge articles.

Software Engineer, Data Systems | NAU SICCS | Flagstaff, AZ **Jan 2025 – Aug 2026**

- Engineered Python-based web scrapers with Selenium and BeautifulSoup to extract unstructured UltraSignUp trail race data.
- Improved scraping throughput by nearly 90% via concurrency and adaptive wait time strategies.
- Reduced OpenStreetMap API calls by 48.5% through dynamic GPX coordinate batching.

Projects

ArtemiS3 | Capstone Project (USGS and NASA) **Svelte | FastAPI | PostgreSQL | Boto3 | Meilisearch**

- Architected a Dockerized search platform enabling discovery across public NASA and USGS AWS S3 datasets.
- Built REST API endpoints with FastAPI and optimized queries using Meilisearch and Boto3.
- Led technical planning and stakeholder presentations, delivering architecture, feasibility, and requirements documentation.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Java, HTML, CSS

Libraries/Frameworks: React, Svelte, AngularJS, Node.js, Next.js, FastAPI, LangChain, LangGraph, PyTorch, Scikit-Learn, Pandas, NumPy

Databases/Search: PostgreSQL, pgvector, SQLite, Firebase Firestore, Meilisearch

Cloud/DevOps: Docker, Kubernetes, LangFuse, LiteLLM, Azure, AWS, GCP, NGINX, CI/CD

Developer Tools: Git, SLURM, Linux, Bash, CMake, UV, Anaconda, Codex, Claude Code, AI-assisted development, Agile